

P2: Defining and Measuring Quality – Worksheet

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Overview

In this task you'll start thinking about the meaning of quality, and reflect on the quality attributes of real software from different perspectives.

Nobody can learn for you!

The point of study is to develop **your** capabilities: your knowledge, skills, and ability to use these effectively.

Working with others (including AI) can be a powerful support in this, but be careful you are not avoiding actual learning. Doing something new is challenging, and it can feel pretty uncomfortable, but stick with it to set yourself up for future success. This will be an important skill in later units and your career.

Do the following:

- Watch the [content videos on Ed](#) for Topic 2. What was the most interesting thing you learned?

> The most interesting thing I learned was... [the systematic thinking as being a software engineer](#) , such that I shall break down each big task into a smaller specific section and record every tiny of them and tick them off as a list.

- Try out the [activities for Topic 2 on Ed](#), and compare your answers with our sample answers. Were your answers different or the same to ours? Why do you think that is?

>

Functional attribute : can the software do the function

Non functional attribute : how well the software can do the job

User perspective :

- 1) Can user login to their account efficiently
- 2) Can user borrow their book smoothly
- 3) Can user pay their outstanding late fees through a comfortable channel

Developer perspective :

- 1) Is the book data storage database correctly configured
- 2) Will the system block out non authorized personal from accessing certain material
- 3) Can user register and use their account

Comparing answer

Teacher's answer is a lot more technical than mine , they have provided a lot of technical terms and words which I'm not included in my previous answer. I looking forward to make my software development journey to be more technical and understandable by other programmer.

Test your new knowledge and skills:

- Briefly explain the concept of quality:

> Quality is...whether a software does what is supposed to do and meet certain user criteria which in the path of development has undergone testcases, internal testing , and well maintained over time.

- Briefly explain the software quality attribute **scalability**, and explain how it is different to **flexibility**. Use an example.

> Scalability : the ability of a system able to handle increased amount of workload without losing performance

Flexibility : the ability of a system able to adapt changes or new requirement without major redesign.

- Pick a piece of software that you use regularly. This could be a website, an app, a game -- literally anything work appropriate. Once you've selected your software:
 - Tell us about it:

> The software I am writing about is...Youtube , it can allows millions of users to watch videos at the same instance and able to push relevant video to the according user . It adapts the new trend of short video which it was originally a long video platform .

- Rank quality attributes from the **developer's** perspective:

> My ranking of quality attributes from the developer's perspective is (rearrange these to form your answer):

1. Security
2. Availability
3. Reliability
4. Portability
5. Maintainability
6. Interoperability
7. Usability
8. Reusability
9. Flexibility
10. Scalability
11. Testability
12. Performance

I chose this ranking because...I think everyone privacy matters , and the software has to able to run which is reliable and available. I find nowadays a lot user from mobile phone so it shall be portable and not bulky. Besides I need to be able to access and understand each part of the software to make it maintainable and interpret to other so I can get some user in.

- Rank quality attributes from the **user's** perspective:

> My ranking of quality attributes from the **user's** perspective is (rearrange these to form your answer):

1. Security
2. Usability
3. Interoperability
4. Portability
5. Availability
6. Reliability
7. Scalability
8. Performance

I chose this ranking because...everyone privacy matters again, and also the software should be able to use. People needs to understand the software before they will start using with a mobile phone. It shall be always available and reliable and fitted to the trends. Performance may be a later story .

- Pick any piece of software you use regularly. Write a small set of tests for that piece of software, documenting them as we have shown you. The tests can be for any part of the software you like, but make sure you write at least 10.

Don't worry

We want you to practice correct test documentation, but we don't expect you to actually find a bug. Most likely, all of your tests will pass.

- Export your test documentation to PDF so you are ready for submission.
- Write a bug report for any bug or unexpected behaviour you have experienced. Make sure you follow the report structure we have shown you.
- Export your bug report to PDF so you are ready for submission.

Be creative

This is not a task about finding bugs, just about practicing how to communicate them. So, your bug could be in software, or in real life – get creative with it! It does not have to be a “real” bug. For example, a fun "bug" report we have seen in the past for this task described how a student's friends did not react to the instagram reels they sent at 1am

Reflect:

- What was your approach to prioritising attributes? Did you find one perspective easier than another?

> My prioritize for youtube ranked : Usability > Availability > Scalability > Performance > Security

I find user perspective is easier than developer perspective : as user perspective is mostly on visual , such as button position , profile , homepage structure

- Why is it so important to understand and define what quality means for a piece of software?

> Defining quality allows developer to understand the whole structure and progressive of a developing software in terms of : performance , security , availability , scalability , usability , and reusability .

Disclose:

- Explain your use of Generative AI for this task. If you didn't use it, you can just write "I did not use any AI for this task". If you did use it, please state which AI model you used (e.g. ChatGPT) and how.

> I use Chatgpt for assistance such as understanding question and improve my thought

Submit:

- Export this worksheet as a PDF.
- Upload your submission to OnTrack